

Project Brief

UFMFV8-15-3: Group Design and Integration Project

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Overview

For your project, you will work in a small group (4 or 5 people) to **design, implement, and evaluate** a 'pick and place' system – a robot arm that can be trained to repeatedly move an item from one location to another. You will be provided with a Lynxmotion AL5D robot arm as a foundational platform, but you must develop your own control system as well as adapting the mechanical components as appropriate meet your project requirements. Each group will choose a different application to work with. You will derive your system requirements from your chosen application, so each group will ultimately develop a different system.



Figure 1: The AL5D robot arm, including controller board.

Scenario

You should imagine your team to be a start-up company working to develop a prototype of a 'minimum viable product' which you will use to attract investment. You will need to develop an outline business model around which to develop your product, and you will need to demonstrate to investors that the product is nearly ready for the market. Hence all core functionality should be proven to be working, with consideration for relevant regulations.

Polished aesthetics and a detailed marketing plan are not required – the investment will be used for that – but your prototype should show due consideration for the user experience (e.g. avoiding excessive bulk or frustrating controls) and financial viability of the concept.

Naturally, you will need to convince the investors that people in your target market will want to buy your system. Hence you must account for users' perspectives from the earliest stages of your project. When gathering and analysing requirements, you should consult with

prospective users. The academic staff on this module will be available throughout the year to play the role of the 'user' in such discussions.

Applications

Your group may propose your own application as the focus of your project, or choose from the following:

Test-Tube Handling

Many processes in chemical/biological laboratories require test tubes to be moved between different devices at specific times. This work can be tedious and, because the human workers need to be highly trained in lab practices, expensive. For this application, your group will design a robot to improve efficiency in such processes by automatically moving test tubes from one rack to another at times set by the user.

Some emerging companies have gone further with this concept, developing the notion of '[Cloud Labs](#)'. You could choose to focus your project towards some variant of this new business model.

Box Packing Robot for Manufacturing

Packing/palletising robots are common in many mass manufacturing environments. However, they still have scope to expand into many sectors, including more specialised producers working at smaller production volumes.

Golf Ball Tee-ing Robot

At a golf driving range, people like to hit a few balls from the same place with minimal interruption. A robot that automatically places balls on the tee for them would allow them to spend more time focussing on their swing.

Key Fetcher for Automated Car Hire Desk

Long queues and restricted office hours are frequent causes of complaints from customers of car hire companies. In this application, your company will supply car hire companies with a robot that automatically fetches the appropriate keys from a rack and drops them in a location where the customer can retrieve them. The car hire company will be able to adapt the robot for each venue by 'teaching' it where the key rack and deposit location are.

Constraints

Group

You will work in small groups, ideally 4-5 students per group. Your groups will be assigned to you, because evidence suggest this provides a more positive experience in various respects. However, if you feel that your assigned group does not provide the most suitable arrangement to support your learning, please raise this with the module staff ASAP so that any reasonable adjustments can be made. You are expected to work with your group to foster a positive collaborative relationship, preventing and resolving disputes in a professional manner.

Application

Your group may choose any of the suggested applications or propose their own, to be discussed with the module staff. Multiple groups can choose the same application, but they must work independently of each other. Your chosen application must include a safety risk to be managed, i.e. some element whose malfunction could cause harm to people, the environment, or property. Obviously this should be discussed with the module staff to ensure the risks associated with the development of your project are acceptable.

Functionality

Your system must include the following functionality:

- Teleoperation: the user must be able to operate the system from a safe distance, to avoid collisions.
- Teach mode: the system must allow a user without specialist skills to 'teach' the device to perform a custom task, which the system can subsequently execute independently.
- Working envelope: radius of at least 50cm from the centre of the robot arm's base to the object to be picked.
- Use of LEDs or other outputs for status/configuration/error signalling.

Domain Specific Constraints

The following sub-sections describe further constraints which vary depending on your group's range of disciplines (which programmes students are enrolled in).

Mechanical

The motors and structural elements of the AL5D robot arm can be included as part of your developed system. You may dismantle and reconfigure the components as necessary to best

meet the requirements of your application. You may also introduce additional off-the-shelf or custom components, subject to the limitations of your project budget.

You must not irreversibly modify any components of the AL5D arm; at the end of the project, you must return it to the module staff, fully assembled in its original configuration.

Extent of modifications required

- *All groups:* At least one addition/replacement of mechanical components is necessary to meet the 50 cm working envelope criterion. Additionally, you must give confidence that your design is capable of its mechanical task across all relevant operating conditions.
- *If you have two or more group members enrolled in Mechanical/Automotive Engineering programmes,* you are required to provide detailed modelling/analyses to give confidence that your design satisfies mechanical requirements (e.g. stress, flexion, fatigue) of relevance to your chosen application. You are also required to produce some additional modification to customise the robot for your application. Suggestions include
 - Customised gripper
 - Peripheral parts, e.g. custom rack for objects to be picked from or placed on.

Electronics

Lynxmotion controller boards may be available for preliminary testing of the motors and mechanical assembly of your robot arm, but these cannot be included as part of your system. You must design/select your own electronics, including power to the motors. You may choose any target platform as your embedded controller, but you must justify your decision.

Custom PCB vs off-the-shelf electronics:

- *If you have three or more group members enrolled in an Electronic Engineering programme (any programme with 'Electronic' in the title),* then you must produce a custom PCB to replace the original controller board.
- *If you have fewer than three electronic engineers in the group,* you can replace the controller boards with suitable off-the-shelf microcontroller/motor driver boards. If you take this route, you will be expected to devote more effort to the software/mechanical aspects of the project.
 - *If you have fewer than two electronic or robotic engineers in the group,* you only need to include the replacement microcontroller and power-supply in your high-level design (i.e. choose and justify appropriate modules and ways for integrating them). You do not need to implement them in your prototype; which can be reliant on the SSC-32u board that comes with arm. You may still need to implement any additional LEDs, sensors, etc on a breadboard, depending on your application.

Software

You may develop your software with whatever language and tools you deem appropriate for the professional application chosen, but you must justify your decisions. External libraries may be used. However, you should thoroughly consider and manage the *technical* and *commercial* risks that such libraries may introduce. Use of a complex external library may necessitate more extensive testing than purpose-built software.

Embedded software:

- *If your group has two or more members enrolled in either Electronics or Robotics programmes, your robot should be capable of working standalone (no PC connection) during automated operation. Hence you will need to include some form of embedded software. This is achievable even if the only programming language you know is MATLAB.*
- *All other groups, you have the option to have the robot reliant on a PC connection at all times, sending messages to/from the board via serial connection.*

Budget

Each group has a budget of £75 to cover prototype development costs, e.g. purchase/production and delivery of components. Extensions to this budget may be possible, subject to prior discussion with the academic staff.

Note that this project budget is distinct from the costing that you will use to justify your business model, which can include economies of scale but should also include a rough estimate of labour costs and any major overheads.

Documenting Your Work

Rigorous systems engineering relies on transparent shared documentation of your work. It is expected that your group will maintain such documentation in a format that all group members and module staff can access easily, e.g. any or all of the following: a shared drive, MS Teams page, or github repository. This documentation should include, but not be limited to:

- Team
 - Team Members
 - Member roles
 - Integration lead
- Plans

- Gantt chart
 - Work breakdown structure
 - Risk register
 - Design requirements (stakeholder, system/high-level, and low-level reqs.)
 - Verification & validation plans
- Logs
 - Meeting minutes
 - Design notes
 - Test logs
- Reference material (links or copies)
 - Datasheets
 - Articles / whitepapers / application notes
- Design implementations
 - Code
 - CAD files

Your group report may refer to this documentation as evidence of your project management process, although it is expected that a concise summary of your group's approach to documentation will be included within the report.